VELOOP REWARDS - Developer Analytics Handoff

Data Analyst Internship, Task 01 | Work Package N

Authenticity notice: Every requirement below is **Proposed** or **To Be Verified**. No event is claimed to exist in VELOOP, and Implementation status is 'To be confirmed by Engineering'. No simulated or real VELOOP statistics appear here.

1. How to use this document

- Read the **general requirements** (Section 2) and **pre-conditions** (Section 3) first. They apply to every event and explain what must exist before events can be trusted.
- Then read the **25 event requirements** (Section 4): 9 Critical and 16 High. Each has a trigger, source, timestamp rule, deduplication key, rules, payload fields, KPI dependency and two testable acceptance tests (one positive, one negative).
- Engineers record their answer in the spreadsheet columns 'Engineering response' (Confirmed existing, Will implement, Needs discussion, Not feasible) and 'Engineering owner'.
- Medium and Low events in the Event Tracking Plan follow the same pattern and can be specified when they are scheduled.

2. General requirements

ID	Requirement	Description	Acceptance test	Priority
AG-01	Common payload on every event	Every event carries event_id, event_name, event_timestamp, user_id, schema_version, platform, app_version and source; session_id when session tracking exists; the pipeline adds received_timestamp.	Sample of events from each type: every one contains all common properties with valid types.	Critical
AG-02	UTC timestamps	All timestamps are ISO 8601 in UTC. Backend outcomes use the server clock, not the device clock.	A change made at a known server time appears with that time in UTC; no event has a local-time offset missing.	Critical
AG-03	Emit after commit	An event is emitted only after the business change is committed. A failed or rolled-back change emits nothing.	Force a failed write: no event exists. Complete a successful write: exactly one event exists.	Critical
AG-04	Retry-safe event_id	A retry of the same event reuses its event_id, so duplicates can be removed.	Send the same event twice: the analytics store holds one.	Critical
AG-05	Analytics never blocks users	Failure to deliver an event must not fail the user's action. Undelivered events are queued and retried.	Disable the analytics endpoint: user actions still succeed and events arrive after it returns.	High
AG-06	No personal data or secrets	No names, contact details, precise location, passwords, tokens, API keys or payment credentials appear in any payload.	Automated scan of sample payloads finds none of these.	Critical
AG-07	Schema versioning	schema_version is set on every event. A breaking change is a new version; existing consumers keep working.	Release a new version: old-version events still pass validation.	High
AG-08	Catalogue names only	Only event names in the Event Tracking Plan are emitted. Unknown names are rejected or quarantined at collection.	Send an unknown event_name: it is quarantined with rule DQ020.	High
AG-09	Environment separation	Test, simulated and production events are kept apart and never mixed.	Events from a test build never appear in the production stream.	Critical
AG-10	Single approved-value source	Enums (statuses, units, methods, categories) are defined once and shared by the app and the pipeline.	Changing an approved value in one place changes validation everywhere.	High
AG-11	Outcomes from the backend	Interactions (opened, viewed) may come from the frontend. Outcomes (completed, credited, reversed, status changes) come from the backend.	Outcome events show source = backend.	High
AG-12	Documented and logged changes	Every released event, property or rule change is recorded in the change log (Work Package Q).	Each change has a log entry with old and new definition.	Medium

3. Pre-conditions

These come from the analytics gaps in the Product Data Audit. Without them, the events below cannot produce trustworthy reporting.

ID	Pre-condition	Gap	Priority	Priority reason	Acceptance test
AP-01	A stable, anonymous, immutable user_id exists for every account and survives device changes.	G01	Critical	Without a stable user_id, one person can appear as several users	The same person on two devices has one user_id; an id is never reused.
AP-02	Every wallet entry has a unique transaction_id, and writes are idempotent.	G02	Critical	Without a stable transaction ID, reliable wallet deduplication cannot be guaranteed	Repeating a request produces one entry; no two entries share a transaction_id.
AP-03	Task attempts are stored with a task_activity_id and the time validation finished.	G03	Critical	All earning analytics depend on completion events	For any completed task there is one attempt row with started, claimed and completed times.
AP-04	Reward units are defined and every reward and wallet entry carries exactly one unit.	G04	Critical	Wrong totals would mislead management	No entry has a missing unit; sums by unit reconcile.
AP-05	Activity and reward are separate records: a reward is credited only after the wallet write succeeds.	G05	Critical	Overstated reward figures	A completed task with a failed wallet write shows no credited reward.
AP-06	Each withdrawal state change is stored with its own timestamp.	G06	High	Without state history, only the latest status is visible	For a completed withdrawal, requested, processing and final times all exist.
AP-07	One activation definition is agreed and stored with a rule version.	G07	High	Inflated referral success	Every activated referral shows the rule version applied.
AP-08	Failures are recorded with a category, not only a generic status.	G09	High	Needed to act on problems, not just count them	Every failed or rejected record has a category from the approved list.
AP-09	The provider's reference id is stored for every externally driven activity.	G10	High	External systems are outside VELOOP control	Provider-validated attempts carry a provider_reference_id.
AP-10	Timestamps are stored in UTC with a known format.	G11	High	Wrong day bucket changes DAU and streaks	No stored timestamp lacks a timezone or has an unreadable format.
AP-11	Product confirms which features are live and which are planned.	G13	High	Wasted engineering effort and empty dashboards	The event list is trimmed to live and imminent features.
AP-12	A data store and event collection route are chosen and documented.	G14	High	Pipeline design depends on it	Events from a test action arrive in the raw store.

Common payload on every event

Field	Type	Required	Meaning
event_id	string	Yes	Unique key; a retry reuses it
event_name	string	Yes	Name from the catalogue
event_timestamp	timestamp (UTC)	Yes	When the fact happened
received_timestamp	timestamp (UTC)	Yes (added by pipeline)	When the pipeline received it
user_id	string	Yes	Stable user key
schema_version	string	Yes	Event definition version
platform	enum	Yes	android, ios, web
app_version	string	Yes	App version that emitted it
source	enum	Yes	frontend, backend, provider
session_id	string	Optional	Present only if session tracking exists

4. Event requirements

AE-001 user_registered | Priority: Critical | Status: To Be Verified

Trigger	Account creation succeeds
Definition	A new user account has been created
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: user_id (one per user)
Event rules	registration_source in approved list; exactly one event per user_id; timestamp not in future
Why needed	Start of lifecycle; acquisition analysis
KPI dependency	Registrations; Cohort Retention (Day N)
Acceptance test (positive)	Given a new account is created and committed, exactly one user_registered exists for that user_id with a registration_source from the approved list.
Acceptance test (negative)	Given account creation fails or is rolled back, no user_registered is recorded.
Notes	Depends on a stable user_id (AP-01).

Payload field (plus common payload AG-01)	Data type	Required	Validation
registration_source	enum	Yes	In approved list; 'unknown' instead of null
registration_method	enum	No	In approved list
referral_code_used	string	No	Matches an existing referral code or null
campaign_id	string	No	In list or null
country_code	string	No	Valid code; never precise location

AE-002 task_started | Priority: Critical | Status: Proposed

Trigger	Task attempt record created
Definition	User began an attempt at a task
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: task_activity_id
Event rules	task_activity_id unique; task_id exists; one event per attempt
Why needed	Denominator for completion rate
KPI dependency	Feature Usage; Feature Adoption Rate; Task Starts; Task Completion Rate; Repeat Usage Rate; Provider Callback Reliability
Acceptance test (positive)	When a task attempt record is created, exactly one task_started exists with a unique task_activity_id.
Acceptance test (negative)	Retrying the same request does not create a second task_started for the same task_activity_id.

Payload field (plus common payload AG-01)	Data type	Required	Validation
task_activity_id	string	Yes	Unique; immutable; generated when the record is created
task_id	string	Yes	Must exist in Tasks
task_type	enum	Yes	In approved list
feature	enum	Yes	In approved list
provider_name	string	No	Required when validation_method = provider
provider_reference_id	string	No	Unique per provider
attempt_number	integer	No	Integer >= 1

AE-003 task_completed | Priority: Critical | Status: Proposed

Trigger	Completion has passed the applicable validation
Definition	A task attempt was validated as completed
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: task_activity_id
Event rules	validation_status = passed; exactly one per task_activity_id; reward_unit in approved list; reward_amount not negative
Why needed	Counts real completions only
KPI dependency	Daily Active Users (DAU); Weekly Active Users (WAU); Monthly Active Users (MAU); Returning Users; Cohort Retention (Day N); Feature Adoption Rate; Task Completions; Task Completion Rate; Help-Center Engagement
Acceptance test (positive)	When a valid task completion occurs, exactly one analytically usable completion event is recorded for the relevant task and user attempt, subject to the final engineering implementation and idempotency design.
Acceptance test (negative)	A completion claim that fails validation produces task_failed and no task_completed.
Notes	Emit when validation passes, not when the user claims completion (AP-03).

Payload field (plus common payload AG-01)	Data type	Required	Validation
task_activity_id	string	Yes	Must exist in Task_Activity
task_id	string	Yes	Must exist in Tasks
task_type	enum	Yes	In approved list
reward_amount	decimal	Yes	Not negative
reward_unit	enum	Yes	Required when amount present
validation_status	enum	Yes	In approved list
reward_id	string	No	Must exist in Rewards
provider_reference_id	string	No	Unique per provider

AE-004 reward_credited | Priority: Critical | Status: Proposed

Trigger	Wallet balance updated with the reward
Definition	A reward has actually been added to the wallet
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: reward_id
Event rules	Exactly one per reward_id; wallet_transaction_id must exist; amount greater than 0
Why needed	Counts rewards actually given
KPI dependency	Reward Transactions; Reward Credits; Reward Reversal Rate; Reward Activity by Feature; Pending Rewards; Net Reward Amount
Acceptance test (positive)	When the wallet balance is updated with a reward, exactly one reward_credited exists carrying the wallet_transaction_id of the credit entry.
Acceptance test (negative)	If the wallet write fails, no reward_credited is recorded.
Notes	Emit only after the wallet write succeeds (AP-05).

Payload field (plus common payload AG-01)	Data type	Required	Validation
reward_id	string	Yes	Must exist in Rewards
reward_amount	decimal	Yes	Not negative
reward_unit	enum	Yes	Required when amount present
source_feature	enum	Yes	In approved list
wallet_transaction_id	string	Yes	Must exist in Wallet_Transactions
task_activity_id	string	No	Must exist in Task_Activity

AE-005 reward_reversed | Priority: Critical | Status: Proposed

Trigger	A credited reward is taken back
Definition	A previously credited reward was reversed
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: original_reward_id (one reversal per credited reward)
Event rules	original_reward_id must refer to a credited reward; no second reversal of the same reward
Why needed	Reward reliability
KPI dependency	Reward Transactions; Reward Reversals; Reward Reversal Rate; Net Reward Amount
Acceptance test (positive)	When a credited reward is taken back, one reward_reversed exists referencing its original_reward_id.
Acceptance test (negative)	A second reversal of the same original reward is refused and produces no second event.
Notes	Original reward is never edited; the reversal links to it.

Payload field (plus common payload AG-01)	Data type	Required	Validation
reward_id	string	Yes	Must exist in Rewards
original_reward_id	string	Yes	Must exist in Rewards (not equal to the record's own id)
reward_amount	decimal	Yes	Not negative
reward_unit	enum	Yes	Required when amount present
reason_category	enum	Yes	No free text
wallet_transaction_id	string	Yes	Must exist in Wallet_Transactions

AE-006 wallet_transaction_created | Priority: Critical | Status: To Be Verified

Trigger	Ledger entry created
Definition	A wallet entry was created
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: transaction_id (and idempotency_key where used)
Event rules	wallet_transaction_id unique; amount greater than 0; direction agrees with transaction_type
Why needed	Every balance change has a record
KPI dependency	Wallet Transactions
Acceptance test (positive)	When a ledger entry is created, one event exists with a unique transaction_id and an amount greater than 0.
Acceptance test (negative)	Repeating a request with the same idempotency_key creates no second entry and no second event.
Notes	An idempotency_key is recommended (AP-02).

Payload field (plus common payload AG-01)	Data type	Required	Validation
wallet_transaction_id	string	Yes	Unique; immutable; generated when the record is created
transaction_type	enum	Yes	In approved list
direction	enum	Yes	Agrees with transaction_type
amount	decimal	Yes	Greater than 0
unit	enum	Yes	One unit per entry
reference_id	string	Yes	Must exist in referenced table
source_feature	enum	Yes	In approved list
related_transaction_id	string	No	Must exist in Wallet_Transactions (not equal to the record's own id)
exchange_group_id	string	No	One debit and one credit

AE-007 wallet_transaction_reversed | Priority: Critical | Status: To Be Verified

Trigger	Reversal entry created
Definition	An earlier wallet entry was reversed
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: transaction_id of the reversal entry
Event rules	related_transaction_id exists and is not itself a reversal
Why needed	Reversal tracking
KPI dependency	Wallet Reversals
Acceptance test (positive)	When a reversal entry is created, one event exists whose related_transaction_id points to an existing original entry.
Acceptance test (negative)	A reversal that points to itself or to another reversal is refused and produces no event.

Payload field (plus common payload AG-01)	Data type	Required	Validation
wallet_transaction_id	string	Yes	Must exist in Wallet_Transactions
related_transaction_id	string	Yes	Must exist in Wallet_Transactions (not equal to the record's own id)
amount	decimal	Yes	Greater than 0
unit	enum	Yes	One unit per entry
reason_category	enum	Yes	No free text

AE-008 withdrawal_requested | Priority: Critical | Status: To Be Verified

Trigger	Withdrawal request created
Definition	User requested a withdrawal
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: withdrawal_id
Event rules	withdrawal_id unique; requested_amount greater than 0; method in approved list
Why needed	Start of withdrawal lifecycle
KPI dependency	Daily Active Users (DAU); Weekly Active Users (WAU); Monthly Active Users (MAU); Returning Users; Cohort Retention (Day N); Feature Adoption Rate; Withdrawal Requests; Pending Withdrawals; Processing Time; Withdrawal Completion Rate; Pending Queue Age; Withdrawal Volume; Withdrawal Method Distribution; Help-Center Engagement
Acceptance test (positive)	When a withdrawal request is stored, one event exists with a method from the approved list and requested_amount greater than 0.
Acceptance test (negative)	A request refused before it is stored produces no withdrawal_requested.
Notes	Whether the balance is held at request time is an open question (see the Wallet and Withdrawal specification).

Payload field (plus common payload AG-01)	Data type	Required	Validation
withdrawal_id	string	Yes	Unique; immutable; generated when the record is created
requested_amount	decimal	Yes	Greater than 0
requested_unit	enum	Yes	In approved list
method	enum	Yes	In approved list
wallet_transaction_id	string	No	Must exist in Wallet_Transactions

AE-009 withdrawal_completed | Priority: Critical | Status: To Be Verified

Trigger	Payout finished
Definition	A withdrawal was paid out
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: withdrawal_id
Event rules	Follows withdrawal_processing; one linked debit; completed_amount not above requested unless a fee rule exists
Why needed	Payout output
KPI dependency	Completed Withdrawals; Pending Withdrawals; Processing Time; Withdrawal Completion Rate; Withdrawal Volume; Withdrawal Method Distribution
Acceptance test (positive)	When a payout finishes, one event exists with completed_amount and the linked wallet_transaction_id.
Acceptance test (negative)	A withdrawal in rejected status never emits withdrawal_completed.

Payload field (plus common payload AG-01)	Data type	Required	Validation
withdrawal_id	string	Yes	Must exist in Withdrawals
method	enum	Yes	In approved list
completed_amount	decimal	Yes	Required when completed
requested_unit	enum	Yes	In approved list
wallet_transaction_id	string	Yes	Must exist in Wallet_Transactions
provider_reference_id	string	No	Unique per provider

AE-010 user_logged_in | Priority: High | Status: Proposed

Trigger	Login succeeds
Definition	A user has started an authenticated visit
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: event_id only (one per login)
Event rules	user_id must exist; timestamp not before registration
Why needed	Return behaviour; visit counts (kept separate from 'active')
KPI dependency	Daily Visitors
Acceptance test (positive)	Given a successful login, exactly one user_logged_in exists for that login and its user_id exists in Users.
Acceptance test (negative)	A failed login attempt produces no user_logged_in.
Notes	Login events may be absent for automatic sessions (A004).

Payload field (plus common payload AG-01)	Data type	Required	Validation
attempt_number	integer	No	Integer >= 1

AE-011 onboarding_started | Priority: High | Status: Proposed

Trigger	First onboarding screen shown
Definition	User has entered the onboarding flow
Source	Frontend
Timestamp	event_timestamp = device time converted to UTC; the pipeline adds received_timestamp so clock drift can be measured.
Deduplication	Primary: event_id (a retry reuses it). Natural key: user_id + onboarding_step (once per onboarding run)
Event rules	onboarding_step in current flow list; once per user unless flow restarts
Why needed	Onboarding funnel entry
KPI dependency	Onboarding Completion Rate
Acceptance test (positive)	When the first onboarding screen is shown to a registered user, one onboarding_started with the onboarding_step is recorded.
Acceptance test (negative)	Re-opening the same step in the same run does not create a second event.
Notes	Onboarding step list to be confirmed.

Payload field (plus common payload AG-01)	Data type	Required	Validation
onboarding_step	string	Yes	Non-empty; no personal data

AE-012 onboarding_completed | Priority: High | Status: Proposed

Trigger	Final onboarding step done
Definition	User has finished onboarding
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: user_id (once per user)
Event rules	Must follow onboarding_started; once per user
Why needed	Activation proxy
KPI dependency	Onboarding Completion Rate
Acceptance test (positive)	When the final onboarding step is completed, one onboarding_completed exists and follows an onboarding_started for the same user.
Acceptance test (negative)	No onboarding_completed exists for a user without an earlier onboarding_started.
Notes	Treatment of skipped onboarding to be agreed.

Payload field (plus common payload AG-01)	Data type	Required	Validation
onboarding_step	string	Yes	Non-empty; no personal data
step_completed_count	integer	No	Whole number, 0 or more

AE-013 account_status_changed | Priority: High | Status: To Be Verified

Trigger	account_status updated by user, system or operations
Definition	An account moved from one status to another
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: user_id + changed_at + new_status
Event rules	previous_status differs from new_status; values in approved status list
Why needed	Track restrictions and closures
KPI dependency	Account Status Mix
Acceptance test (positive)	When account_status changes from A to B, one event exists with previous_status = A, new_status = B and the actor.
Acceptance test (negative)	An update that leaves the status unchanged produces no event.
Notes	Real status names must be mapped (A019).

Payload field (plus common payload AG-01)	Data type	Required	Validation
previous_status	enum	Yes	Differs from new_status
new_status	enum	Yes	In approved list
status_change_actor	enum	Yes	In approved list
reason_category	enum	No	No free text

AE-014 task_failed | Priority: High | Status: Proposed

Trigger	Attempt ends without validated completion
Definition	A task attempt failed or was abandoned
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: task_activity_id
Event rules	failure_category in approved list; cannot follow task_completed for the same attempt
Why needed	Explain drop-off and failures
KPI dependency	Task Failure Mix
Acceptance test (positive)	When an attempt ends failed or abandoned, one task_failed exists with a failure_category from the approved list.
Acceptance test (negative)	No task_failed exists for a task_activity_id that has a task_completed.

Payload field (plus common payload AG-01)	Data type	Required	Validation
task_activity_id	string	Yes	Must exist in Task_Activity
task_id	string	Yes	Must exist in Tasks
failure_category	enum	Yes	Required when failed or abandoned
failure_detail_code	string	No	No personal data
provider_name	string	No	Required when validation_method = provider
provider_reference_id	string	No	Unique per provider

AE-015 reward_pending | Priority: High | Status: Proposed

Trigger	Reward created but not yet usable
Definition	A reward is held awaiting validation or confirmation
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: reward_id
Event rules	reward_id unique; reward_amount greater than 0
Why needed	Pending reward queue
KPI dependency	Reward Transactions; Pending Rewards
Acceptance test (positive)	When a reward is created but not yet usable, one reward_pending exists with a pending_reason.
Acceptance test (negative)	A reward credited immediately produces no reward_pending.

Payload field (plus common payload AG-01)	Data type	Required	Validation
reward_id	string	Yes	Unique; immutable; generated when the record is created
reward_amount	decimal	Yes	Not negative
reward_unit	enum	Yes	Required when amount present
source_feature	enum	Yes	In approved list
pending_reason	enum	Yes	Required when pending
task_activity_id	string	No	Must exist in Task_Activity

AE-016 daily_reward_claimed | Priority: High | Status: Proposed

Trigger	Daily reward successfully claimed
Definition	User claimed the daily reward
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: user_id + claim_date
Event rules	One per user per claim_date; reward_amount not negative
Why needed	Daily habit measurement
KPI dependency	Daily Active Users (DAU); Weekly Active Users (WAU); Monthly Active Users (MAU); Returning Users; Cohort Retention (Day N); Feature Adoption Rate; Streak Length; Help-Center Engagement
Acceptance test (positive)	When a user claims the daily reward, one event exists with claim_date and streak_day.
Acceptance test (negative)	A second claim on the same claim_date produces no second event.

Payload field (plus common payload AG-01)	Data type	Required	Validation
claim_date	date	Yes	ISO 8601 date
streak_day	integer	Yes	Whole number, 0 or more
reward_amount	decimal	Yes	Not negative
reward_unit	enum	Yes	Required when amount present
reward_id	string	No	Must exist in Rewards

AE-017 referral_created | Priority: High | Status: Proposed

Trigger	Referral code or link created
Definition	A referral was created by a user
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: referral_id
Event rules	referral_id unique; referring_user_id must exist
Why needed	Referral funnel start
KPI dependency	Referrals Created; Referral Conversion Rate
Acceptance test (positive)	When a referral record is created, one referral_created exists with a unique referral_id.
Acceptance test (negative)	A referral with a non-existent referring_user_id is refused and produces no event.

Payload field (plus common payload AG-01)	Data type	Required	Validation
referral_id	string	Yes	Unique; immutable; generated when the record is created
referring_user_id	string	Yes	Must exist in Users
referral_code_used	string	No	Matches an existing referral code or null
campaign_id	string	No	In list or null

AE-018 referral_registered | Priority: High | Status: Proposed

Trigger	Referred user completes sign-up
Definition	A referred user registered
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: referral_id
Event rules	referred_user_id differs from referring_user_id; follows referral_created
Why needed	Registration is not conversion
KPI dependency	Referral Registrations; Referral Activation Rate
Acceptance test (positive)	When a referred user registers, one referral_registered exists and referred_user_id differs from referring_user_id.
Acceptance test (negative)	A self-referral produces no referral_registered.

Payload field (plus common payload AG-01)	Data type	Required	Validation
referral_id	string	Yes	Must exist in Referrals
referring_user_id	string	Yes	Must exist in Users
referred_user_id	string	Yes	Must exist in Users

AE-019 referral_activated | Priority: High | Status: To Be Verified

Trigger	Referred user meets activation rule
Definition	A referred user became activated
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: referral_id
Event rules	Follows referral_registered; once per referral_id
Why needed	Quality of referrals
KPI dependency	Referral Activations; Referral Activation Rate
Acceptance test (positive)	When the activation rule is met, one referral_activated exists with the activation_rule_version applied.
Acceptance test (negative)	No referral_activated exists for a referral without an earlier referral_registered.
Notes	The activation rule is not yet agreed (A010, AP-07).

Payload field (plus common payload AG-01)	Data type	Required	Validation
referral_id	string	Yes	Must exist in Referrals
referred_user_id	string	Yes	Must exist in Users
activation_rule_version	string	Yes	Matches change log

AE-020 referral_converted | Priority: High | Status: Proposed

Trigger	Referral meets success condition
Definition	A referral was successful
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: referral_id
Event rules	Follows referral_activated unless rule says otherwise; once per referral_id
Why needed	True referral success
KPI dependency	Referral Conversions; Referral Conversion Rate
Acceptance test (positive)	When the success condition is met, one referral_converted exists for that referral_id.
Acceptance test (negative)	A second conversion for the same referral_id produces no event.

Payload field (plus common payload AG-01)	Data type	Required	Validation
referral_id	string	Yes	Must exist in Referrals
referring_user_id	string	Yes	Must exist in Users
referred_user_id	string	Yes	Must exist in Users
reward_id	string	No	Must exist in Rewards
wallet_transaction_id	string	No	Must exist in Wallet_Transactions

AE-021 wallet_transaction_completed | Priority: High | Status: To Be Verified

Trigger	Ledger entry becomes final
Definition	A wallet entry was completed
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: transaction_id
Event rules	Follows wallet_transaction_created; once per transaction
Why needed	Settlement and balance integrity
KPI dependency	Wallet Transactions; Credit Volume; Redemption Activity; Wallet Reconciliation Exceptions
Acceptance test (positive)	When an entry becomes final, one completed event exists and follows the created event for the same transaction_id.
Acceptance test (negative)	No completed event exists for an entry that failed.

Payload field (plus common payload AG-01)	Data type	Required	Validation
wallet_transaction_id	string	Yes	Must exist in Wallet_Transactions
transaction_type	enum	Yes	In approved list
direction	enum	Yes	Agrees with transaction_type
amount	decimal	Yes	Greater than 0
unit	enum	Yes	One unit per entry
transaction_status	enum	Yes	Value in the approved list

AE-022 withdrawal_processing | Priority: High | Status: To Be Verified

Trigger	Request enters review or processing
Definition	A withdrawal started processing
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: withdrawal_id
Event rules	Follows withdrawal_requested; once per withdrawal
Why needed	Processing time and queue
KPI dependency	Pending Withdrawals; Pending Queue Age
Acceptance test (positive)	When review or processing begins, one event exists and follows withdrawal_requested for the same withdrawal_id.
Acceptance test (negative)	No second processing event is recorded for the same withdrawal_id.

Payload field (plus common payload AG-01)	Data type	Required	Validation
withdrawal_id	string	Yes	Must exist in Withdrawals
method	enum	Yes	In approved list
provider_reference_id	string	No	Unique per provider

AE-023 withdrawal_rejected | Priority: High | Status: To Be Verified

Trigger	Request refused
Definition	A withdrawal was rejected
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: withdrawal_id
Event rules	reason_category required; cannot follow withdrawal_completed
Why needed	Understand friction
KPI dependency	Pending Withdrawals; Rejected Withdrawals; Processing Time
Acceptance test (positive)	When a request is refused, one event exists with a reason_category.
Acceptance test (negative)	No withdrawal_rejected exists for a withdrawal that is already completed.

Payload field (plus common payload AG-01)	Data type	Required	Validation
withdrawal_id	string	Yes	Must exist in Withdrawals
method	enum	Yes	In approved list
reason_category	enum	Yes	No free text

AE-024 withdrawal_reversed | Priority: High | Status: To Be Verified

Trigger	Completed payout undone or returned
Definition	A completed withdrawal was reversed
Source	Backend
Timestamp	event_timestamp = server time when the change was committed (UTC, ISO 8601). received_timestamp is added by the pipeline.
Deduplication	Primary: event_id (a retry reuses it). Natural key: withdrawal_id
Event rules	Must follow withdrawal_completed
Why needed	Payout reliability
KPI dependency	Reversed Withdrawals
Acceptance test (positive)	When a completed payout is reversed, one event exists with a reason_category.
Acceptance test (negative)	No withdrawal_reversed exists for a withdrawal that never completed.

Payload field (plus common payload AG-01)	Data type	Required	Validation
withdrawal_id	string	Yes	Must exist in Withdrawals
method	enum	Yes	In approved list
reason_category	enum	Yes	No free text
wallet_transaction_id	string	Yes	Must exist in Wallet_Transactions
provider_reference_id	string	No	Unique per provider

AE-025 provider_callback_received | Priority: High | Status: To Be Verified

Trigger	External provider reports a result (name to be agreed)
Definition	A provider callback arrived
Source	Provider
Timestamp	event_timestamp = provider_event_time if supplied, otherwise callback_received_at; both are stored (UTC).
Deduplication	Primary: event_id (a retry reuses it). Natural key: provider_name + provider_reference_id + provider_status
Event rules	provider_reference_id present; stored separately from VELOOP status
Why needed	Reliability of external providers
KPI dependency	Provider Callback Reliability
Acceptance test (positive)	When a provider callback arrives, one event exists storing provider_reference_id, provider_status and callback_received_at.
Acceptance test (negative)	A repeated callback with the same provider_reference_id and status creates no second event.
Notes	Event name to be agreed.

Payload field (plus common payload AG-01)	Data type	Required	Validation
provider_name	string	Yes	Required when validation_method = provider
provider_reference_id	string	Yes	Unique per provider
provider_status	enum	Yes	Value in the approved list
callback_received_at	timestamp	Yes	ISO 8601 UTC
provider_event_time	timestamp	No	ISO 8601 UTC
task_activity_id	string	No	Must exist in Task_Activity

5. Open questions

- Which of these events exist today, and which service would emit each? (A001, A011)

- Which approved lists (statuses, units, methods, categories) will be used? (A002, A006)
- Which idempotency design will be used for wallet and event writes? (A005)
- What retry and queue approach is acceptable for undelivered events?
- Which onboarding steps, account statuses and failure categories are real? (A003, A019)

Next: Work Package O - Data Pipeline and Operating Model.